

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

II Semester of B. Sc. (Biological Sciences) Examination, May 2018

Course code & Name: BS 210 & Analytical Techniques in Biology

Date: 03/05/2018 Day: Thursday Time: 10:00 A.M. To 10:30 A.M. Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.	20
1.	An elution process in which the composition of the mobile phase is changed by mixing two solvents during a chromatographic separation is known as _____ elution.	01
	(i) Isocratic (ii) Isotopic	
	(iii) Gradient (iv) Denaturing	
2.	The following phenomenon is exhibited by a molecule, which absorbs the electromagnetic radiation in the UV range and emits radiations in the visible range.	01
	(i) Reflection (ii) Diffraction	
	(iii) Scattering (iv) Fluorescence	
3.	The following blotting technique is used for detection of specific RNA fragments by hybridization to a labelled gene probe.	01
	(i) Western blotting (ii) Eastern blotting	
	(iii) Southern blotting (iv) Northern blotting	
4.	The following is the basic principle of all types of chromatographic separations.	01
	(i) Retardation factor (R_f) (ii) Partition coefficient (K_d)	
	(iii) Absorbance (iv) All of them	

5.	A biological particle will exhibit a zero sedimentation rate upon reaching the _____ position in the density gradient centrifugation.		01
	(i) Ionic	(ii) Isoelectric	
	(iii) Isopycnic	(iv) All of them	
6.	In bacterial genomic DNA isolation, the following detergent is used for removing lipid molecules and thereby causing disruption of the cell membranes.		01
	(i) SDS	(ii) EDTA	
	(iii) Lysozyme	(iv) Proteinase K	
7.	The following blue colored dye is used to track the progress of electrophoretic migration in agarose gel electrophoresis.		01
	(i) Blue dextran	(ii) Bromophenol blue	
	(iii) Commasie brilliant blue	(iv) All of them	
8.	A DNA preparation is quantified at _____ wavelength in a UV-Spectrophotometer, at this wavelength one absorbance unit corresponds to 50 µg/ml of double-stranded DNA.		01
	(i) 525 nm	(ii) 280 nm	
	(iii) 260 nm	(iv) 340 nm	
9.	The following chromatography separates biomolecules on the basis of their molecular size and shape.		01
	(i) Ion-exchange	(ii) Gel-filtration	
	(iii) Affinity	(iv) All of them	
10.	The following step in the PCR allows the hybridisation of the two oligonucleotide primers to their complementary sites that flank the target DNA.		01
	(i) Annealing	(ii) Extension	
	(iii) Denaturation	(iv) All of them	
11.	The following is the most commonly used method for concentration of DNA samples.		01
	(i) Ethanol precipitation	(ii) Ammonium sulphate precipitation	
	(iii) DNAase treatment	(iv) Phenol-chloroform extraction	
12.	The best resolution that can be achieved by a conventional light microscope is___ µm.		01
	(i) 0.5	(ii) 0.0005	
	(iii) 0.4	(iv) 0.2	
13.	Of the following, which electromagnetic radiation has a wavelength shorter than visible light?		01
	(i) Microwave	(ii) Ultraviolet	
	(iii) Infrared	(iv) Radiowaves	

14.	The _____ microscope utilizes doughnut shaped electromagnets called magnetic lenses as objectives.		01
	(i) Fluorescence	(ii) Phase contrast	
	(iii) Electron	(iv) Compound	
15.	The phase contrast microscope converts slight differences in the _____ and cell density within the sample into easily detected variations in light intensity.		01
	(i) Refractive index	(ii) Fluorescence	
	(iii) pH	(iv) All of them	
16.	Cation exchanger stationary phase possesses _____ functional groups which will attract cationic analytes.		01
	(i) Neutral	(ii) Positively charged	
	(iii) Basic	(iv) Negatively charged	
17.	For concentration of proteins in a sample, adding more amount of ammonium sulfate salt would eventually decrease the solubility of proteins and promote their precipitation, this is known as _____.		01
	(i) Salting in	(ii) Salting out	
	(iii) Desalting	(iv) All of them	
18.	Resolution of ascending paper chromatography is _____ descending technique since two opposite forces act on the solute: the capillary action, which makes it move up and the gravitational force which opposes this movement.		01
	(i) Less than	(ii) Similar to	
	(iii) More than	(iv) All of them	
19.	In SDS-PAGE, the following reagent binds strongly to, and denatures, the proteins and confers net negative charge on them.		01
	(i) EDTA	(ii) SDS	
	(iii) Ethanol	(iv) APS	
20.	The Beer-Lambert law states that the absorbance of a solution is directly proportional to _____ of the absorbing molecules in the solution and the path length.		01
	(i) Precipitation	(ii) Denaturation	
	(iii) Concentration	(iv) Oxidation	

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

II Semester of B. Sc. (Biological Sciences) Examination, May 2018

Course code & Name: BS 210 & Analytical Techniques in Biology

Date: 03/05/2018 Day: Thursday Time: 10:30 A.M. To 1:00 P.M. Maximum Marks: 50

Instructions:

1. Section I and II must be attempted in TWO ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I		
Q – II Answer the following questions as directed (answer any four).		20
1.	Explain the principle and working of an epifluorescence microscope.	05
2.	Describe the principle and instrumentation of a UV-visible spectrophotometer.	05
3.	Explain the optical arrangements and working of a phase contrast microscope.	05
4.	Explain the principle and geometry of optical arrangements in a spectrofluorimeter.	05
5.	Describe the agarose gel electrophoresis method for separation of DNA fragments.	05
6.	Describe the principle and elution methods of affinity chromatography.	05
SECTION - II		
Q-III Answer the following questions as directed (answer any six).		30
1.	Explain the various steps for isolation of genomic DNA from bacteria.	05
2.	Give an account of the generalized approach used for protein purification.	05
3.	Write a short note on polymerase chain reaction.	05
4.	Explain the principle of following: (a) Ion-exchange chromatography (b) Gel-filtration chromatography	05
5.	Explain the various components of a gas chromatography system.	05
6.	Explain the southern blotting technique.	05
7.	Describe the different forms of paper chromatography.	05
8.	Give an account of the major types of laboratory centrifuges encountered by undergraduate students.	05